

MUHAMMAD AMMAD

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SOFTWARE ENGINEER

linkedin.com/in/ammad-mughal/

Full-Stack Software Engineer with **4+ years** of experience building scalable, responsive, and secure web applications. Proficient in **Next.js, React, TypeScript, Python, and Flask**. Skilled in **Web and Mobile** application development utilising **TDD** and **CICD pipeline** through **Github Actions, RESTful API** development, **database integration**, and **system architecture**.

Experienced in **integrating AI-powered features** using **OpenAI APIs** and machine learning models to enhance user experience and automate workflows. Passionate about delivering modern, high-impact solutions by combining web technologies and AI.

WORK EXPERIENCE

FULLSTACK ENGINEER (FREELANCE)

SELF-EMPLOYED

May 2025 - till date

- Developed **Smart Calling Assistant** using Next.js, CISCO Webex SDK for PSTN calling and Livekit based AI agent with STT, TTS and response generation capabilities.
- Developed **Mediconnect** - a patient appointment automation system using Retell AI API in Next.js along with cronjob to scheduling calls using Node-Cron.
- Built **BestCoverLetter** - an AI-based SaaS platform using OpenAI APIs, NextAuth, Prisma, and Tailwind. Made it multi-lingual using i18n in Next.js.
- Developed multi-lingual product website for **MinaClass** - a learning app - using Webflow.
- Delivered full-stack solution including **web-portal and mobile app** for a local e-commerce hardware business using Next.js and React Native Expo.
- Updated website of **Ntirwaa Catering** - a catering company - developed in Wordpress.

FRONT-END ENGINEER

BROADPEAK TECHNOLOGIES

April 2025 - May 2025

- Contributed to Naranga franchise platform modernization—migrating from monolith to microservices. Developed modern UI with React, RTL, Jest, Redux, and Tailwind CSS.
- Collaborated closely with backend and DevOps teams for **REST API integration**.
- Prepared a UX-driven rollout plan before project pause due to Lahore branch closure.

FRONT-END ENGINEER

ENERGY AND SECURITY GROUP (LLC)

November 2022 - March 2025

- Developed a **role-based CRM portal** using React, Redux, Storybooks, SCSS, and Material UI to digitize manual consumer services—reducing processing time from months to hours.
- Implemented **paperless workflows**, reducing carbon footprint and operational overhead.
- Integrated with legacy billing systems** to eliminate data duplication and improve accuracy.
- Built a **real-time complaint tracking dashboard** with Google Maps API and KMZ layers.
- Created a **monitoring dashboard** showing live feeder statuses, transformer data, and geo-tagged complaints.
- Optimized performance using **lazy loading, throttling, debouncing**, and Redux Toolkit.
- Secured the portal with **JWT-based authentication, CAPTCHA, SQL injection protection, and rate limiting**.

SOFTWARE ENGINEER

POWER INFORMATION TECHNOLOGY COMPANY

March 2020 - October 2022

- Developed a **Store Inventory System (SIS)** with Flask, Python, and Jinja2, featuring secure JWT APIs and full CRUD operations.
- Re-engineered **IBS and Power ERP**, creating BPMN/UML diagrams and system documentation.
- Deployed **self-hosted web solutions** (Power Meetings, Power Chat, GitLab Server, Help Desk) on Ubuntu and CentOS.

EDUCATIONAL HISTORY

BACHELOR IN COMPUTER SCIENCE
3.08 CGPA
INTERMEDIATE (PRE-ENGINEERING)
77% Marks
MATRICULATION (SCIENCE)
78% Marks

GCU, LAHORE
October 2015 – December 2019
GCU, LAHORE
March 2012 – May 2014
GCMS, LAHORE
January 2010 – February 2012

TECHNICAL SKILLS

- **Front-End:** Next.js, React, Redux, Tailwind CSS, TypeScript, Storybooks, RTL and Jest
- **Back-End:** Python, Flask, RESTful APIs, Web-Sockets, Next-Auth
- **Databases:** PostgreSQL, SQL Server, MongoDB, Supabase, Prisma, Mongoose
- **AI & Automation:** OpenAI API, Whisper, LangChain, AI-powered custom agents
- **Web Security:** JWT Auth, reCAPTCHA, API Rate Limiting
- **Tools & Platforms:** CI/CD through Github Actions, Docker, Jira, Slack, Vercel, AWS

KEY PROJECTS

- **AI Calling Assistant (Ongoing):** Developing an AI-powered voice assistant using Webex Calling SDK, LiveKit Client, Livekit based custom AI agent, in Next.js capable of handling real-time conversations and performing automated call tasks.
- **BestCoverLetter (SaaS Platform):** Developed an AI-based SaaS web app using Next.js, NextAuth, Prisma, PostgreSQL, and Tailwind CSS and using OpenAI APIs to generate personalized cover letters with dynamic templates and user accounts.
- **SPM:** An indigenous, role based project management portal developed for a corporate company utilizing *Flask*, *SQLAlchemy*, *Marshmallow* and *Postgresql* at the backend and *React*, *Tailwind CSS* and *Redux Toolkit* on front-end.
- **e-Operations:** A CRM solution developed in Power Sector Improvement Activities (PSIA), served as a role-based interface for field operations developed using *React*, *Redux Toolkit*, *SCSS* and *Material UI*.
- **MEPCO Smart Web Portal:** A web portal developed as part of the Power Sector Improvement Activities (PSIA) using *React*, *SCSS* and *Material UI* for 33 million power consumers across Pakistan.
- **Kippax:** Developed a sleek, modern, and responsive website for a dynamic marketing agency. Built using *React*, *Tailwind CSS* and *Framer Motion* and *Loveable AI*, the site reflects the agency's creative brand with a bold, clean design and smooth user experience.
- **Store Inventory System (SIS):** Web app built with *Flask*, *Python*, *JavaScript* and *Jinja2*, supporting full process of a standard inventory management business flow along with CRUD for items, categories, and suppliers, hierarchical stores.
- **Chatiniti:** A full-featured real-time chat application using *Flask*, *Flask-SocketIO*, and *PostgreSQL*. Chatiniti supports one-to-one and group chat, along with audio/video calls using WebRTC.

CERTIFICATIONS

Microsoft Certified - Azure Fundamentals

Microsoft

Artificial Intelligence & Computing (AIC)

PIAIC

Foundation of Project Management

Google - Coursera

Project Initiation - Starting a Successful Project

Google - Coursera

IBM Blockchain Essentials

IBM Developer Skills Network

IBM BLOCKCHAIN DEVELOPER

IBM Developer Skills Network

Overview of Datacenter Physical Infrastructure

Schneider Electric